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1. Yeast Strains and Plasmids

1.1 Strains

Strains are detailed in Table S1. The parent strain was YAS245-5C (*can1::HO-CAN1 ho::HO-ADE2 ura3 ade2 leu2 trp1 his3*) and is a W303a derivative¹. We constructed ACLY379 in two steps. First we replaced the *BARI* coding sequence with the *URA3* gene by PCR-mediated one-step replacement² using pRS406 as a template. Then, we selected spontaneous *ura3* mutants by plating the cells on 5-FOA (5-fluoro-orotic-acid, Sigma-Aldrich Inc, St. Luis, MO) plates³. We constructed ACLY387 by replacing the *PRM1* coding sequence in ACLY379 with a PCR product containing the entire *YFP* coding sequence, the *ADHI* terminator and the *S.pombe his5⁺* gene from pYFP-His3MX6¹. We checked all PCR-mediated integrations by colony PCR to verify the structure of the expected junctions. We constructed ACLY394 from ACLY387 by replacing the endogenous *CDC28* gene with the mutant copy carried by the plasmid pCDC28-as2-406 (*cdc28-F88A*) using the two step integration-excision gene replacement approach³. We integrated into the *trp1* genomic locus in ACLY394 the plasmid pPRM1-CFP-404 to make TCY3096 and the plasmid pACT1-CFP-404 to make TCY3154. We constructed GPY3262 and GPY3263 by deleting *FUS3* and *KSSI* from TCY3154 using PCR mediated one-step deletion with the pFA6a-kanMX6 template⁴.

We made the strains with all different binary combinations of the constitutive promoters (P_{STE5} , P_{BMH2} , P_{ACT1} and P_{PRP22}) driving GFP derivatives (XFP) by integrating two different P_{XXX} -XFP plasmids into different locations in the genome of ACLY379. We first transformed ACLY379 with a P_{XXX} -XFP plasmid bearing the *URA3* marker and then with a second P_{XXX} -XFP plasmid bearing the *TRP1* marker. We constructed TCY3146, TCY3148 and TCY3204 as follows. First we integrated into the *ura3* genomic locus of ACLY379 a plasmid with a constitutive promoter driving YFP, then integrated

into the *leu2* locus the plasmid pPRM1-CFP-405, and lastly replaced CDC28 with the *cdc28-as2* allele present in pCDC28-as2-404 by two-step integration-excision using 5-FAA (5 Flouoro-anthranilic acid) to select the excision products ⁵.

After transformation of a strain with an integrating fluorescent reporter plasmid, we grew cells from single colonies and observed by quantitative fluorescence that individual transformants often had different mean fluorescence intensities, consistent with some isolates having multiple copies of the integrated plasmid. We observed that there was a distribution of XFP fluorescence in cells grown from single colonies. After subtracting autofluorescence and calculating the mean of this distribution for each tested colony, we found that colonies differed by integer multiples of the lowest mean value, suggesting that colonies with the lowest mean value had one copy and those with twice this value had two copies and so on. This lowest mean value depended on the identity of the promoter driving the XFP. Therefore, in each step of strain construction, we discarded strains with more than one copy integrated in the genome and selected for further use those that had a single copy integrated in the genome as suggested by the mean value of fluorescence.

1.2 Plasmids

We constructed YFP and CFP reporter plasmids as follows. We subcloned a BglII fragment that contained the *GAL1* promoter from pACL7 ¹ into pRS406 ⁶ to make pGAL406. We amplified by PCR the CEN-ARS from pRS416 using primers that introduced terminal AatII restriction sites, and ligated the resulting product into AatII-cut pGAL406 to make pTC7. We made YFP and CFP coding sequences by PCR mediated site-directed mutagenesis of the *GFP* sequence from *A. victoria* and used them to replace the *GFP* sequence in pTC7 by co-transformation and recombination in yeast to make pTC7-YFP and pTC7-CFP. We removed the *GAL1* promoter fragment to make a plasmid backbone into which we introduced ~1 kb fragments containing the regulatory regions

found upstream of the start codons for *PRM1*, *STE5*, *BMH2*, *ACT1*, and *PRP22*, here called "promoters". Then we removed the AatII CEN-ARS fragment to convert the URA3 marked plasmids into integrating vectors. To make *TRP1* and *LEU2* marked vectors with the same P_{XXX}-XFP inserts, we subcloned BglII fragments from these *URA3* marked vectors into BglII cut pRS404 (*TRP1*) or pRS405 (*LEU2*) plasmids. We then integrated the reporter plasmids into the desired chromosome location by transformation with plasmids linearized within the locus homologous to the genomic target locus. Additional details are available upon request. To construct pCDC28-as2-406 and pCDC28-as2-404, we performed PCR-mediated site directed mutagenesis using genomic DNA from ACLY379 cells (which have a wild type *CDC28* gene) as a template to introduce the F88A mutation and terminal NotI and XhoI sites. The fragment thus generated started at nucleotide 4 of the *CDC28* coding sequence and ended 393 bases downstream of the stop codon and it included a mutated codon 88 (TTT -> CGT). We then cloned this mutant *cdc28-as2* allele into the pRS406 and pRS404 vectors cut with NotI and XhoI.

2 Single Cell Microscopy

We maintained cultures in exponential growth for at least 15 hours in BSM medium (BIO-101, Inc.) with 2% glucose before microscopy. We sonicated the cells to disperse clumps and added 100 µl of cell suspension (roughly 10⁶ cells) to 96-well glass bottom plates that had been pre-coated with conA (concanavalin A type V, Sigma-Aldrich). To coat with conA, we added to the well 100 µl of a 100 µg/ml solution of conA in water. We incubated the plate for 1 hour at RT and then washed 3 times with water. We kept coated wells in water for up to 2 days before adding cells. After settling and binding for 10 minutes, we washed away unbound cells. We used a 60X PlanApo objective (N.A.=1.4) under oil immersion in a Nikon TE2000 inverted microscope equipped with a mercury lamp, a motorized XYZ stage and 512BFT MicroMax cooled CCD camera (Photometrix, Tucson AZ) and located in a 30°C room. We automatically and repeatedly

imaged multiple wells over time using MetaMorph 5.0 software (Universal Imaging Corporation, Downingtown, PA), supplemented with custom made software (Cell-ID 1.0, see Supp Mat Part2/Gordon et al. in prep) that enabled autofocus and subsequent image analysis. We also used MetaMorph to control the built-in motorized Z-motor and cube changer, the XY-motorized stage (MS2000, Applied Scientific Instrumentation, Eugene, OR), and brightfield and fluorescence light shutters (Uniblitz, Rochester, NY). We manually selected three or more image fields per well, acquired time 0 images, and then changed to medium with α factor (Sigma-Aldrich) and/or the 1-NM-PP1 inhibitor (4-amino-1-(tert-butyl)-3-(1'-naphthylmethyl)pyrazolo[3,4-*d*]pyrimidine, Cellular Genomics, Inc, Branford, CT). We found that α factor adsorbs to plastic and glass surfaces when in solution. Therefore, we prepared α factor in BSM medium containing 20 μ g/ml of blocking reagent (casein from the DIG nucleic acid detection kit, Roche Diagnostics Corporation, Indianapolis, IN) to block non-specific adsorption. Blocking non-specific adsorption was the key to obtaining reproducible results when performing experiments with low doses of α factor. We prepared 10 mM stock solutions of 1-NM-PP1 in DMSO and diluted directly into medium to 10 μ M before using. The software captured images every 15 minutes. For each time and field, we acquired a brightfield image and YFP and CFP fluorescence images using filter sets 31044v2 and 41028 from Chroma Technologies Corp. (Brattleboro, Vermont).

3 Calculation of Total Fluorescence

To calculate “system output” of signalling response (defined as the total corrected fluorescence signal in a cell from the corresponding reporter gene) we first subtracted from the “raw” total fluorescence for a given cell (the sum of the fluorescence values of all the pixels that Cell-ID associated with that cell) the background signal measured

outside the cells. We then corrected this value for photobleaching. We also scaled it by a quantity that depended on the cell size and the measured point spread function to correct for the effect of out-of-focus regions of the cell. Moreover, for time courses, we refined the fluorescence measurement by correcting for small fluctuations between time frames in the number of pixels that Cell-ID associated with a given cell. These corrections, together with the measurement of the maturation and stability of the GFP derivatives, are discussed in Gordon et al. (Nature Methods, submitted).

We were concerned that difference in the maturation rates for YFP and CFP might cause significant spurious decorrelation between observed YFP and CFP fluorescence in the experiments described in the paper. We tested this possibility by repeating the experiments described below, but applying cycloheximide to the cells 2.5 hours after α factor stimulation. This treatment blocked all further protein production. We then waited several hours before observation to allow all the fluorophores to mature. We observed that in these conditions, YFP and CFP were not degraded (i.e., fluorescence three hours after cycloheximide addition was the same as after 24 hours). Therefore, data obtained in this way were independent of maturation and degradation rates of YFP and CFP. We attained identical results (not shown) to those described in the paper.

4 Defining Pathway Output and Expression Capacity

We considered the production of YFP or CFP in a cell, where any given promoter drives the YFP or CFP gene. We considered the amount of fluorescent protein produced to be the product of two subsystems. The first subsystem we call “*pathway*”. For the case of an α factor inducible promoter, the input to pathway is the dose of α factor, and the output is the activation of the inducible promoter that drives the fluorescent protein. Pathway output depends on the summed activity of upstream DNA bound transcription factors, from the time of addition of α factor, ΔT . As an example, we examine the case that α

factor leads to a constant level of pathway activation. In this case the pathway output will increase linearly with ΔT since the output is the integral of the level of activation over ΔT . The average of the level of activation is a measure of the pathway capacity of the cell and is given by the total output divided by ΔT . For a given cell i , we describe pathway output as $P_i \Delta T$ where P_i is the time-average of the level of pathway activation (the total output divided by ΔT). We separate P_i as $P_i = L_i + \lambda_i$ where L_i is the expectation value of P_i for a given level of α factor, and λ_i is the stochastic fluctuation term for that cell. We refer to L_i as the pathway power. For the case of an α factor inducible promoter, the pathway “power” is a function of the dose of α factor. We refer to cell-to-cell differences in $P_i \Delta T$ as “variation in pathway output”, and we describe differences that result from the stochastic fluctuation term λ_i as “transmission noise.”

The expectation value L_i is the result we would expect for cell i , in the absence of any stochastic fluctuations in the workings of the pathway for that cell. One can imagine a thought experiment, in which we repeat the same experiment on the same cell i many times. In that analysis, the average of the many experimental trials on cell i would be L_i , while any differences between experimental trials are the result of stochasticity. In the lab, however, we are unable to repeat the same experiment on the same cell, but conceptually we can treat the output of the cell as being decomposed into an expectation value, that depends on the number and activity of the molecules that comprise the pathway, and a stochastic fluctuation that occurs because of the inherent randomness of the activity of the pathway as well as its composition over the time of the experiment. Different cells have different pathway capacities and therefore can have different values of L_i .

The fluorescence that we observe at a given time is the accumulated fluorescence from the time of induction. In the case of an α factor inducible promoter, induction starts soon after addition of α factor. Therefore, the pathway output that we observe will be the

total output during the time period from induction to observation. For constitutive promoters, we treat all the events that lead to the activity of the constitutive promoter as the constitutive pathway. In our data, we subtract the fluorescence observed at the time of induction, and thus, as for the α factor inducible promoter, the pathway output is the total output during the time period from induction to observation.

The second subsystem consists of the sequence of events from gene transcription through protein translation. We call this subsystem “*expression*.” It includes transcriptional initiation, elongation, mRNA maturation, nuclear export, and mRNA translation and degradation. We use our data to measure cell-to-cell variation in expression, but we cannot isolate, with our current data, the contribution of each part of the expression machinery to the overall variation. This is the reason that we have grouped together all the processes from transcriptional initiation through protein expression. It is possible that in the future we will have data that will be sensitive to the contributions of the different parts of the expression machinery, in which case it will be useful to appropriately subdivide the expression subsystem.

The output of the expression subsystem is the total amount of the fluorescent reporter protein, and the input is the level of promoter activity integrated from the beginning of the experiment ($P\Delta T$, the output of the pathway subsystem). As discussed in the main text, we assume that the expression per unit of input is independent of the level of input. For cell i , we describe the expression per unit input as the variable E_i where E_i is given by the sum of G_i and γ_i . The quantity G_i is the expectation value of the expression per unit input for cell i , and this quantity may vary from cell to cell. We refer to G_i as the “expression capacity” for cell i , and we refer to the cell-to-cell differences in G_i as “variation in expression capacity”. The quantity γ_i is the stochastic fluctuation that occurred in cell i , and we refer to differences that result from this stochastic fluctuation term as “expression noise.”

Using this model, we described the total amount of YFP, y_i , in cell i as

$$y_i = (L_i + \lambda_i) \Delta T \times (G_i + \gamma_i) \quad (1)$$

where ΔT is the time from addition of α factor. This equation holds even for the extreme case that the cellular apparatus for gene expression is already saturated at the time of α factor stimulation (for example, in the case that there are few free ribosomes for further translation). In the case of ribosome saturation, the gene expression level may be very small. In any case, our data suggested that the cells in our experiments were not near this saturation point. We examined cells that had the α factor sensitive *PRM1* promoter driving YFP and had the α factor insensitive *ACT1* promoter driving CFP (strain TCY3154). The mean and standard deviation of the CFP signal were indistinguishable for all induction levels (Figure S1). We concluded from this that we were able to change the induced load on the gene expression apparatus by activating the α factor pathway (~ 50 genes are directly induced by α factor ⁷) without affecting a constitutive promoter, and therefore, that the ability of the cells to express genes into proteins is not saturated.

5 Extracting Pathway and Expression Capacity Information from the Data

5.1 Variance and Covariance

We define the variance of a given quantity x for a population of cells as

$$\sigma^2(x) = \frac{1}{N} \sum (x_i - \bar{x})^2 \quad (2)$$

$$= \overline{(x - \bar{x})^2} \quad (3)$$

where N is the number of cells in the population. We use the overbar symbol ($\bar{x}$) to represent the population average. Similarly we define the covariance, $Cov(x,y)$, of two quantities x and y as

$$Cov(x, y) = \frac{1}{N} \sum_i (x_i - \bar{x})(y_i - \bar{y}) \quad (4)$$

$$= \overline{(x - \bar{x})(y - \bar{y})} \quad (5)$$

The covariance will be non-zero if x and y are correlated (or anti-correlated). For example, if whenever x_i is above $\bar{x}$, y_i is also above its average, then the covariance will be positive.

The correlation coefficient, $\rho(x, y)$, is the covariance scaled by the standard deviations. Specifically

$$\rho(x, y) \equiv \frac{Cov(x, y)}{\sigma(x)\sigma(y)} \quad (6)$$

5.2 Calculating Variance and Covariance of G , γ , L , and λ

As discussed above and in the main body of the paper, at a given level of α factor, the average pathway output (P) for a given cell has a pathway power L associated with it, along with a stochastic term, λ . Thus, we have $P = L + \lambda$. Similarly, the expression subsystem of the cell (E) contains a capacity term G and a stochastic term γ .

For a given level of α factor, the amount of fluorescent reporter protein (y_i) produced for a given cell i is the product of the terms

$$y_i = (L_i + \lambda_i) \Delta T \times (G_i + \gamma_i) \quad (7)$$

$$= (L_i G_i + L_i \gamma_i + \lambda_i G_i + \lambda_i \gamma_i) \Delta T \quad (8)$$

Since the expectations of the stochastic fluctuation terms are zero, and since their fluctuations are uncorrelated to other terms, we expect that the population average of y_i reduces to the population average of the quantity $L_i G_i \Delta T$.

To calculate the average of y , we re-write the product $L_i \times G_i$ in terms of population averages and deviations from the average. For these two terms, we use a capital delta (Δ) to represent a deviation from a population average. We get

$$L_i = \bar{L} + \Delta L_i \quad (9)$$

$$G_i = \bar{G} + \Delta G_i \quad (10)$$

We then calculate the population average of y to be

$$\bar{y} = \frac{\Delta T}{N} \sum_i (L_i \times G_i) \quad (11)$$

$$= (\bar{L} \times \bar{G} + \text{Cov}(L, G)) \Delta T \quad (12)$$

where we used equation (4) for the definition of covariance and the fact that $\overline{\Delta G} = \overline{\Delta L} = 0$. Thus the average number of YFP molecules is the product of $\bar{L}$ and $\bar{G}$ plus an extra term to account for their correlation.

To calculate the variance on y , we re-write equation (7) in terms of equations (9) and (10). We get

$$y_i = (\bar{L} + \Delta L_i + \lambda_i) \Delta T \times (\bar{G} + \Delta G_i + \gamma_i) \quad (13)$$

$$= \left(1 + \frac{\Delta L_i}{\bar{L}} + \frac{\lambda_i}{\bar{L}}\right) \times \left(1 + \frac{\Delta G_i}{\bar{G}} + \frac{\gamma_i}{\bar{G}}\right) \times \bar{L} \times \bar{G} \times \Delta T \quad (14)$$

$$= \left(1 + \frac{\Delta L_i}{\bar{L}} + \frac{\lambda_i}{\bar{L}}\right) \times \left(1 + \frac{\Delta G_i}{\bar{G}} + \frac{\gamma_i}{\bar{G}}\right) \times \bar{y} \times \left(1 - \frac{\text{Cov}(L, G) \Delta T}{\bar{y}}\right) \quad (15)$$

$$\approx \bar{y} \times \left(1 + \frac{\Delta L_i}{\bar{L}} + \frac{\lambda_i}{\bar{L}} + \frac{\Delta G_i}{\bar{G}} + \frac{\gamma_i}{\bar{G}}\right) \quad (16)$$

where we used equation (12) to make the substitution for $\bar{L} \times \bar{G} \times \Delta T$ in going from equation (14) to (15); and where we have dropped, in going from equation (15) to (16) all

higher order terms of fractional deviations from the means. These higher order terms are various products of $(\Delta L_i / \bar{L})$, $(\lambda_i / \bar{L})$, $(\Delta G_i / \bar{G})$, and $(\gamma_i / \bar{G})$. We assume that each of the higher order terms is small relative to the lower order terms that are retained in equation (16), and therefore we have neglected them. The term $Cov(L, G) \Delta T / \bar{y}$ is second order and has also been neglected. We discuss the magnitude of the error introduced by this approximation in section 7.1 below.

We use equation (16) to calculate the variance on the number of YFP molecules using the definitions of variance and covariance above. We get

$$\frac{\sigma^2(y)}{\bar{y}^2} = \frac{\sigma^2(L)}{\bar{L}^2} + \frac{\sigma^2(\lambda)}{\bar{L}^2} + \frac{\sigma^2(G)}{\bar{G}^2} + \frac{\sigma^2(\gamma)}{\bar{G}^2} + 2\rho(L, G) \frac{\sigma(L)}{\bar{L}} \frac{\sigma(G)}{\bar{G}} \quad (17)$$

where $\rho(L, G)$ is the correlation coefficient between L and G .

The correlation coefficient is always between -1 and 1, and a value of 0 corresponds to no correlation. There may be no correlation between L and G , but, for example, one possibility that would lead to a correlation is that cells with higher expression capacity may have stronger or weaker pathway output, perhaps because they have higher amounts of positive regulators or negative regulators of the pathway, respectively. This would show up as a positive or negative value for $\rho(L, G)$, respectively.

We do not expect that the stochastic fluctuations λ or γ are correlated to any other variables, and thus the terms $\rho(L, \lambda)$, $\rho(L, \gamma)$, $\rho(\gamma, G)$, $\rho(\lambda, G)$, and $\rho(\lambda, \gamma)$ were omitted from equation (17).

All the variances in equation (17) are expressed as fractions of the means squared. We define the variables

$$\eta(L) \equiv \sigma(L) / \bar{L} \quad (18)$$

$$\eta(\lambda) \equiv \sigma(\lambda) / \bar{L} \quad (19)$$

$$\eta(G) \equiv \sigma(G) / \bar{G} \quad (20)$$

$$\eta(\gamma) \equiv \sigma(\gamma) / \bar{G} \quad (21)$$

The quantity η is simply the width of different distributions expressed as a fraction of their means. For the stochastic variables λ and γ , we will refer to $\eta(\lambda)$ and $\eta(\gamma)$ as the “noise” associated with those quantities, and for the non-stochastic variables L and G , we will refer to $\eta(L)$ and $\eta(G)$ as the “variation.”

We quantified cell-to-cell variation in system output using “normalized variance” ($\eta^2 = \sigma^2/\mu^2$) rather than “noise strength”⁸ (σ^2/μ), a measure others have used to describe deviations from purely stochastic Poisson-type biological processes⁹. We used normalized variance for a number of reasons. First, we found that most of the cell-to-cell differences in system behavior we reported are not due to stochastic differences in signal transmission or gene expression, as described in the main text. Second, use of η^2 allowed examination of different amounts of variation in terms of the fraction of the mean. Third, and most important, because η^2 is unitless, it allowed direct comparison of different measurements, for example, from different GFP derivatives, and it allowed the definition of total variation as the sum of individual sources of variation plus additional terms to account for correlations.

We re-write equation (17) in terms of η as

$$\eta^2(y) = \eta^2(P) + \eta^2(G) + \eta^2(\gamma) + 2\rho(L, G)\eta(L)\eta(G) \quad (22)$$

We have written $\eta^2(P)$ for $\eta^2(L) + \eta^2(\lambda)$ since we are considering the case that α factor is constant for all the cells and since the data will not be able to distinguish the cell-to-cell variation in pathway output from the noise or stochastic fluctuations. The term $\eta^2(P)$ is the variation in average pathway output per unit time, which, because pathway output is

given by $P\Delta T$ and ΔT is the same for every cell, is identical to variation in pathway output.

To separate $\eta(G)$ from $\eta(P)$ and $\eta(\gamma)$ in the data from the two-promoter, two-color experiments, discussed further below, we introduce the quantity

$$Z(y,c) = \frac{\eta^2(y) + \eta^2(c)}{2} - \eta(y)\eta(c)\rho(y,c) \quad (23)$$

The quantity Z is the average variance divided by the mean square of the two fluorescence signals YFP (y) and CFP (c) with the correlated part subtracted out. It thus is a measure of the uncorrelated part of the y vs c scatter plot. $Z(y,c)$ will be used explicitly to calculate $\eta(\gamma)$ and $\eta(P)$ from the data, as discussed below.

5.3 Two Color Variants Driven by the Same Promoter and Gene Expression Noise

We calculate the correlation coefficient between YFP and CFP for the case that both genes are expressed in the same cell and with the same promoter as

$$\rho(y,c) \equiv \frac{1}{\sigma(y)\sigma(c)} \frac{1}{N} \sum_i \Delta(y_i)\Delta(c_i) \quad (24)$$

$$= \frac{1}{\eta(y)\eta(c)} \frac{1}{N} \sum_i \left(\frac{\Delta L_i}{L} + \frac{\lambda_i}{L} + \frac{\Delta G_i}{G} + \frac{\gamma_{y,i}}{G} \right) \times \left(\frac{\Delta L_i}{L} + \frac{\lambda_i}{L} + \frac{\Delta G_i}{G} + \frac{\gamma_{c,i}}{G} \right) \quad (25)$$

where we have used equation (16) for the deviations from the mean, $\Delta(y_i)$, and an analogous equation for $\Delta(c_i)$. Since the YFP gene has the same promoter as CFP, ΔL_i is the same for both color variants. The quantity λ_i is the same since stochastic fluctuations in pathway output occur upstream of the promoters (with the exception of the binding of the transcription factors to each individual copy of the reporter genes, see the end of this section), and the quantity ΔG_i is the same since the proteins are being expressed in the same cell. The stochastic fluctuations in gene expression, γ_i , however, are different for the two color variants and we write them as $\gamma_{y,i}$ and $\gamma_{c,i}$ for YFP and CFP respectively.

The uncorrelated terms drop out of equation (25) when we perform the population average. We do not expect the stochastic fluctuations to be correlated with any other terms, and equation (25) becomes

$$\rho(y,c) = \frac{\eta^2(P) + \eta^2(G) + 2\rho(L,G)\eta(L)\eta(G)}{\eta(y)\eta(c)} \quad (26)$$

where $\eta^2(P) = \eta^2(L) + \eta^2(\lambda)$ as discussed above.

The quantity $Z(y,c)$ as defined in equation (23), gives the contribution of the uncorrelated part of the expression of the two color variants to the average η^2 of the two colors. Only the gene expression noise is uncorrelated, and we get

$$Z(y,c) = \frac{\eta^2(\gamma_y) + \eta^2(\gamma_c)}{2} \quad (27)$$

$Z(y,c)$ is the average of gene expression noise of the two promoters. Since the same promoters are driving the YFP and CFP genes, we expect equal levels of mRNA for the two variants on average, and therefore the gene expression noise is the same for the YFP and CFP signal. We get $\eta^2(\gamma_y) = \eta^2(\gamma_c) \equiv \eta^2(\gamma)$, and therefore, from equation (27), $Z(y,c) = \eta^2(\gamma)$.

Figure S4 shows plots of $Z(y,c) = \eta^2(\gamma)$ for several combinations of constitutive promoters (*ACT1*, *BMH2* and *STE5*) and for the α factor responsive promoter *PRM1*. Interestingly, a single curve seems to describe all the data. The value of $\eta^2(\gamma)$ seems to be a function only of the total number of fluorophores produced, regardless of the promoters responsible. In particular, *PRM1*, the pheromone responsive gene, produces the same level of gene expression noise whether we do a high pheromone exposure for short times, or a low pheromone exposure for long times. This is what we would expect if each mRNA were transcribed and translated independently of the others.

The data fit well to the function $A+B/\mu$ where μ is the median YFP signal. We get the fit results $A=0.0015 \pm 0.0001$, and $B=7000 \pm 500$. This fit is consistent with a Poisson process (since B is greater than zero) plus a constant term (A) that is a minimum level of de-correlation between the YFP and CFP signals. We discuss the possibility that the constant term (A) might be explained as an experimental artifact in section 7.3 below.

We note that for the experiment described above, CFP and YFP were not driven by the same promoter, but rather by identical *copies* of the same promoter. Therefore, stochasticity in the binding of transcription factors to the DNA and in the subsequent activation of transcription contributes to the decorrelated part of the total variation. In our *ad hoc* model, these molecular steps are part of the pathway subsystem, and therefore fluctuations in this step should contribute to the transmission noise, $\eta^2(\lambda)$. However, due to experimental limitations of the type of experiment we have just described, the noise caused by these molecular steps is, instead, included in the measure of gene expression noise $\eta^2(\gamma)$. Our data suggested, though, that this extra stochastic component is an insignificant contributor to $\eta^2(\gamma)$. We concluded this from our results that gene expression noise is a function of the mean output, independent of the promoter that drove transcription of YFP and CFP. Since 1) different promoters are activated by different transcription factors that bind different DNA sequences and 2) different transcription factors might be expected to have different binding and activation kinetics, we expect that noise due to stochasticity in the binding of different transcription factors and noise due to the stochasticity of the transcriptional activation of different promoters would also be different. But we observed that the measured $\eta^2(\gamma)$ is independent of the nature of the promoter. Therefore, we concluded that these extra steps cause an insignificant contribution to the measured gene expression noise.

5.4 Two Color Variants Driven by Different Promoters and Pathway Variation

For this case, the calculation of $\rho(y, c)$ is the same as in equation (25) above except that the pathway activity is different for the two color variants since different promoters are driving the YFP and CFP genes. If we assume that the two terms L and λ are uncorrelated for the two promoters then we get

$$\rho(y, c) = \frac{\eta^2(G) + \rho(L_y, G)\eta(L_y)\eta(G) + \rho(L_c, G)\eta(L_c)\eta(G)}{\eta(y)\eta(c)} \quad (28)$$

where we have written L_y and L_c for capacities of the pathways that lead to the activation of the YFP and CFP gene respectively. The uncorrelated part of the average η^2 will now include a contribution from the pathway; and the quantity $Z(y, c)$ includes this extra contribution. We get

$$Z(y, c) = \frac{\eta^2(P_y) + \eta^2(P_c)}{2} + \frac{\eta^2(\gamma_y) + \eta^2(\gamma_c)}{2} \quad (29)$$

where $\eta^2(\lambda_y)$ is the transmission noise for the pathway that leads to YFP activation and $\eta^2(P_y) = \eta^2(L_y) + \eta^2(\lambda_y)$, and similarly for $\eta^2(\lambda_c)$ and $\eta^2(P_c)$.

Since we could infer the gene expression noise part of equation (29) from data with identical promoters, we can gather enough data to calculate the transmission noise if we consider all pairwise combinations of three different promoters and six data sets. We defined each data set according to which promoter drove which color variant:

Data set	Promoter driving YFP	Promoter driving CFP
1	ACT1	BMH2
2	ACT1	STE5
3	BMH2	STE5
4	ACT1	ACT1
5	BMH2	BMH2
6	STE5	STE5

The last three data sets had identical promoters and so could be used to calculate the gene expression noise for each promoter, as discussed in section 5.3 above. We used these numbers to subtract out the gene expression noise for the first three data sets. We wrote Z_1 for $Z(y,c)$ for data set 1, and similarly for the other data sets. The gene expression noise for ACT1 was given by Z_4 , for BMH2 by Z_5 , and for STE5 by Z_6 . We then used the data to calculate the three quantities A_1 , A_2 and A_3 where we defined

$$\begin{aligned}
A_1 &\equiv Z_1 - \frac{1}{2}(Z_4 + Z_5) = \frac{1}{2} \times [\eta^2(P_{ACT1}) + \eta^2(P_{BMH2})] \\
A_2 &\equiv Z_2 - \frac{1}{2}(Z_4 + Z_6) = \frac{1}{2} \times [\eta^2(P_{ACT1}) + \eta^2(P_{STE5})] \\
A_3 &\equiv Z_3 - \frac{1}{2}(Z_5 + Z_6) = \frac{1}{2} \times [\eta^2(P_{STE5}) + \eta^2(P_{BMH2})]
\end{aligned} \tag{30}$$

This is a linear equation for the η^2 terms, which we solve to give

$$\begin{aligned}
\eta^2(P_{ACT1}) &= A_1 + A_2 - A_3 \\
\eta^2(P_{BMH2}) &= A_1 - A_2 + A_3 \\
\eta^2(P_{STE5}) &= -A_1 + A_2 + A_3
\end{aligned} \tag{31}$$

We exposed strains containing the six combinations of reporter genes to 20 nM pheromone and observed the accumulation of fluorescence for two and a half hours. For each data set and condition, we calculated Z from equation (23). The results were

$$\begin{aligned}
 Z(ACT1, BMH2) &= 0.0045 \pm 0.005 \\
 Z(ACT1, STE5) &= 0.015 \pm 0.003 \\
 Z(BMH2, STE5) &= 0.016 \pm 0.003 \\
 Z(ACT1, ACT1) &= 0.0020 \pm 0.0003 \\
 Z(BMH2, BMH2) &= 0.0045 \pm 0.0005 \\
 Z(STE5, STE5) &= 0.0268 \pm 0.0043
 \end{aligned} \tag{32}$$

Equations (30) and (31) then gave

$$\begin{aligned}
 \eta^2(P_{ACT1}) &= 0.0013 \pm 0.0046 \\
 \eta^2(P_{BMH2}) &= 0.0013 \pm 0.0065 \\
 \eta^2(P_{STE5}) &= 0.0006 \pm 0.008
 \end{aligned} \tag{33}$$

for the variation associated with the pathway part of the total system output of the three constitutive promoters.

Our results thus indicated that the values of pathway variation of the constitutive promoters are small compared to the total variations, which are ~ 0.15 to ~ 0.20 . For each case the pathway variation was statistically consistent with 0. We also observed consistent results before and after α factor stimulation as well as at different pheromone concentrations, as would be expected for α factor insensitive promoters.

For experiments in which the PRM1 promoter drove YFP and the ACT1 promoter CFP, we calculated $Z(y,c)$ and used equation (29) to determine $\eta^2(P_{PRM1})$. To do this, we first multiplied the quantity $Z(y,c)$ for this experiment by two. We then subtracted the predicted gene expression noise for the YFP signal using the median of the YFP signal and the curve shown in Figure S4. We used an analogous curve to subtract the predicted gene expression noise for the CFP signal. As shown by equation (29), the resulting quantity is the sum of the two pathway variations $\eta^2(P_{PRM1}) + \eta^2(P_{ACT1})$. We then used the number $\eta^2(P_{ACT1}) = 0.0035 \pm 0.0025$ to calculate $\eta^2(P_{PRM1})$. This value for $\eta^2(P_{ACT1})$ is

statistically consistent with the result of equation (33) and is the result of an experiment using 5 nM pheromone (not shown).

After the above analysis was complete, we devised a test using two new strains. The first strain had the constitutive promoter *PRP22* driving both YFP and CFP, and the second strain had *PRP22* driving YFP and *ACT1* driving CFP. We then calculated the quantity A as defined in equation (30) as $A = Z(\text{PRP22}, \text{ACT1}) - [Z(\text{PRP22}, \text{PRP22}) + Z(\text{ACT1}, \text{ACT1})]/2$. This quantity is the average of the pathway variations for *PRP22* and *ACT1*. Our results gave $\frac{1}{2} \times [\eta^2(P_{\text{ACT1}}) + \eta^2(P_{\text{PRP22}})] = 0.010 \pm 0.005$ after two and half hours at 20 nM of pheromone. We calculated the same quantity for 2.5 nM pheromone, and the result was consistent with a value of 0 for $\eta^2(P_{\text{PRP22}})$ while the 20 nM result is consistent with a higher value. For both cases, the value of $\eta^2(P_{\text{PRP22}})$ is small compared to the total variation seen in the data, as we also observed for the other constitutive promoters.

Finally, as noted in Section 5.3, our measure of gene expression noise included the contribution of stochasticity in binding of transcription factors. Therefore, measurement of the variation in pathway output described here includes the cell-to-cell variation in pathway power, but the stochastic contribution to the measurement only includes processes up to, but not including, the stochasticity in the binding of the transcription factor to the DNA.

5.5 Expression Capacity Variation and Correlation

We could calculate the total variation, $\eta^2(y)$ of a given fluorescent reporter from the data. The total variation is the sum of several different contributions as described in equation (22). Once we had calculated $\eta^2(\gamma)$ and $\eta^2(P)$, as discussed in sections 5.3 and 5.4, we could subtract them from the total variation. Specifically

$$\eta^2(y) - \eta^2(P) - \eta^2(\gamma) = \eta^2(G) + 2\rho(L, G)\eta(L)\eta(G) \quad (34)$$

The left hand side is determined from data, and the right hand side gave the expression capacity variation plus a correlation term.

Similarly, we could use the data to calculate the average variation of the YFP and CFP data for the case that different promoters drove the two colors. We have the equation

$$\frac{\eta^2(y) + \eta^2(c)}{2} - Z(y, c) = \eta^2(G) + \rho(L_y, G)\eta(L_y)\eta(G) + \rho(L_c, G)\eta(L_c)\eta(G) \quad (35)$$

The left side is determined from the data, and the right side gives the expression capacity variation plus correlation terms for each promoter. Column 7 of Table 1 of the main text reports values for the left side of equation (34).

6 Calculating Means, Variance, and Covariance from the Data

6.1 Median and MAD

For the calculation of population averages, we used the median (MED) of the distributions. For Gaussian distributions the median of a data set is an unbiased estimator of the mean of the population and is robust against outliers¹⁰. Similarly a robust estimator of the standard deviation of a distribution is the median of the absolute deviations from the population median, divided by 0.6745 (MAD) (dividing by 0.6745 gives the correct result for a normal distribution)¹⁰. Specifically, to calculate the MAD, we first calculate the absolute value of the difference between every point and the population median. The median of this new distribution divided by 0.6745 is what we refer to as the MAD.

For asymmetric distributions, the median will not produce the same value as the mean. As a test we drew random values according to a Γ -distribution (see section 7.1 below). The Γ -distribution is bounded at the low end by zero and has a high end tail. We found that the median produced values near the peak of the distribution, while the mean

of the distribution is higher than the peak because of the tail. Similarly the value for MAD was lower than the standard deviation of the distribution. Interestingly, however, the ratio MED/MAD was close to the value of the mean over the standard deviation. Moreover, our data do not show large high end tails, and thus if they do have large tails, the sample sizes are too small to make them pronounced. We concluded that using MED/MAD for an estimator of η did not produce any significant error in our measurements.

6.2 Biweight Midcovariance (BIWT)

For a two-dimensional data set $\{x_i, y_i\}$, the statistic known as the “biweight midcovariance” (BIWT) is a robust estimator of the covariance $\text{Cov}(x, y)$ ¹¹. The BIWT is defined as

$$BIWT = \frac{N \sum a_i (X_i - M_x)(1 - U_i^2)^2 b_i (Y_i - M_y)(1 - V_i^2)^2}{[\sum a_i (1 - U_i^2)(1 - 5U_i^2)][\sum b_i (1 - V_i^2)(1 - 5V_i^2)]} \quad (36)$$

where N is the number of data points, M_x and M_y are the medians of the x - and y -distributions respectively; U_i is the quantity $(x_i - M_x)/9/\text{MAD}(x)$ and V_i is the analogous quantity for the y variable; and a_i is 1 if U_i is between -1 and 1 and 0 otherwise, and similarly for b_i . The sum that defines the BIWT is similar to the definition of covariance except that the U , V , a , and b terms weight the points as a function of their distance from the median. We have used test cases to confirm that our implementation of the BIWT does indeed produce an unbiased estimator of the covariance. We used the BIWT as an estimator of $\text{Cov}(x, y)$ and then calculated the correlation coefficient between YFP and CFP for our data as $\rho(YFP, CFP) = \text{Cov}(YFP, CFP) / (\sigma(YFP) \sigma(CFP))$ where σ is the standard deviation as calculated using the MAD statistic described above.

6.3 Minimum Covariance Determinant (MCD)

As a test, we also implemented another robust method to calculate the means and covariance matrix of the data. The MCD algorithm¹² uses the definition of variance and covariance of equations (2) and (4) above to calculate the covariance matrix but only uses a fixed fraction of the population. The points that are discarded are those that are most likely to be outliers. For our data, we chose the fixed fraction to be 90%. The algorithm started by choosing 90% of the data at random and using those points to calculate the covariance matrix. All the data were then sorted based on the values of the distance $\vec{A} \cdot \tilde{C}^{-1} \cdot \vec{A}$ where $\vec{A}$ was the vector $(YFP - \overline{YFP}, CFP - \overline{CFP})$ for a given cell, and $\tilde{C}^{-1}$ was the inverse of the measured covariance matrix. Cells with larger values of the distance were more likely to be outliers, and the top 10% were ignored, and the covariance matrix was recalculated. This procedure was repeated until the ignored cells at the next iteration did not change.

For time courses, we kept track of the cells that were left out at each time point (thus omitting from the analysis cells for which we did not have a complete time series). We then threw away all cells that were ignored at any time point and recalculated the covariance matrix for each time. For this second iteration, however, we used 100% of the cells that were retained. This allowed us to avoid discrete changes when a cell that was ignored in one calculation at one time point was used in the next. The output of the algorithm are robust estimators of the five quantities, $\overline{YFP}$, $\overline{CFP}$, $\sigma^2(YFP)$, $\sigma^2(CFP)$, and $Cov(YFP, CFP)$.

In the analyses presented, we used the BIWT statistic and the median and MAD for our data; however, the values produced by the MCD algorithm agreed well with those quantities.

7 Treatment of Measurement Errors

7.1 Error Introduced by Dropping Higher Order Terms in Variance Calculations

We arrived at equation (16) after making the approximation that we could neglect the higher order terms in equation (15) as discussed above. We refer to this approximation as the “linearization approximation.” The approximation is good to the extent that the terms $\eta(L)$, $\eta(G)$, $\eta(\lambda)$, and $\eta(\gamma)$ are “much less” than 1. However, we have determined from our data that $\eta(G)$ is relatively large, ~ 0.45 , and therefore we revisit the approximation here. In this section we describe our calculation of the effect of this approximation. As discussed below, we found that the approximation has a negligible effect on the measurements of the cell-to-cell variation in expression capacity, but that the approximation results in a systematic error in the measured pathway variation and gene expression noise of between $\sim 10\%$ and $\sim 15\%$.

The linearization approximation affects our results since equations (22), (27), and (29) (which describe the measured quantities $\eta^2(c)$ and $\eta^2(y)$ and $Z(y,c)$ in terms of our model parameters) were all derived from equation (16), after we made the approximation. To make this discussion simpler, we re-write these equations as follows:

$$\eta^2(y) = \eta^2(L_y) + \eta^2(\lambda_y) + \eta^2(G) + \eta^2(\gamma_y) + 2\rho(L_y, G)\eta(L_y)\eta(G) \quad (37)$$

$$\eta^2(c) = \eta^2(L_c) + \eta^2(\lambda_c) + \eta^2(G) + \eta^2(\gamma_c) + 2\rho(L_c, G)\eta(L_c)\eta(G) \quad (38)$$

$$Z(y, c) = \begin{cases} \frac{\eta^2(\gamma_y) + \eta^2(\gamma_c)}{2} & , \text{ same promoters} \\ \frac{\eta^2(L_y) + \eta^2(\lambda_y) + \eta^2(L_c) + \eta^2(\lambda_c)}{2} + \frac{\eta^2(\gamma_y) + \eta^2(\gamma_c)}{2} & , \text{ different promoters} \end{cases} \quad (39)$$

where equation (38) is the analogue to equation (37) for the CFP gene; where we have added explicit subscripts to the model parameters for the CFP and YFP genes; and where we have written out the term $\eta^2(P)$ as $\eta^2(L) + \eta^2(\lambda)$. We calculate the left hand side (LHS) of each equation from data, and we calculate the right hand side (RHS) from the model. The LHS has no approximation associated with it since it is calculated from data, but the

RHS is only correct to the extent that the linearization approximation introduces negligible errors.

To measure the errors introduced into the RHS of these equations by the approximation, we used a Monte Carlo simulation method. First, we chose distributions for the model parameters G , L_y , and L_c (that determine the non-stochastic part of the cell-to-cell variation) as well as model parameters γ_c , γ_y , λ_c , and λ_y (that determine the stochastic, noise contribution to the cell-to-cell variation). The distributions that we used are discussed below. We then generated random values for these model parameters based on these chosen distributions. We used every set of random values $\{G_i, L_{y,i}, L_{c,i}, \gamma_{c,i}, \gamma_{y,i}, \lambda_{c,i}, \lambda_{y,i}\}$ to determine the behavior of a simulated “cell.” We calculated the YFP signal (y_i) and CFP signal (c_i) of this simulated cell i according to

$$y_i = (L_{y,i} + \lambda_{y,i}) \times (G_i + \gamma_{y,i}) \quad (40)$$

$$c_i = (L_{c,i} + \lambda_{c,i}) \times (G_i + \gamma_{c,i}) \quad (41)$$

where equation (41) is the analogue to equation (40) for the CFP gene, and (40) is simply equation (1) with a subscript y added to indicate the YFP gene. The term ΔT that appears in equation (1) is a constant that we have dropped because it does not affect the η^2 values. The quantity G_i is the randomly chosen expression capacity for cell i ; the quantities $L_{c,i}$ and $L_{y,i}$ are the pathway powers that we randomly chose for cell i (for the CFP and YFP genes respectively); the quantities $\gamma_{c,i}$ and $\gamma_{y,i}$ are the randomly chosen contributions from the gene expression noise; and $\lambda_{c,i}$ and $\lambda_{y,i}$ are the randomly chosen contributions from the transmission noise.

We generated 10,000 random “cells” in this fashion. We then used the generated values for y and c to calculate $\eta^2(y)$, $\eta^2(c)$, and $Z(y,c)$ using the methods of sections 6.1 and 6.2 above. In this way we treated the generated cells as if they were real data, and there is no linearization approximation made in calculating these three quantities. If the

linearization approximation is a good approximation, then these three quantities, $\eta^2(y)$, $\eta^2(c)$, and $Z(y,c)$, should be described well by equations (37), (38), and (39) respectively. Since we chose the distributions for the model parameters, we were able to calculate the right hand sides of these equations. By comparing the right hand side and the left hand sides, we were able to determine the error introduced by the linearization approximation.

We chose distributions for L and G that satisfied two requirements. The first requirement was that L and G had to be always positive, and the second was that the distributions had only a single peak. We made use of gamma (Γ) distributions for this purpose. There are other distributions that also satisfy these requirements, but we reproduced the shape of our data reasonably well with Γ distributions, and so we think they provided a good test of the linearization approximation. The Γ distribution has two parameters, α and θ , and its probability function is given by

$$p(x) = \frac{x^{\alpha-1} e^{-x/\theta}}{\Gamma(\alpha) \theta^\alpha} \quad (42)$$

(<http://mathworld.wolfram.com/GammaDistribution.html>) for $x > 0$, where $\Gamma(\alpha)$ is the gamma function evaluated at α . The mean (μ) of the distribution is given by $\alpha\theta$ and the variance (σ^2) by $\alpha\theta^2$; so that $\eta^2 = \sigma^2/\mu^2$ is given by $1/\alpha$. We therefore chose the two parameters, α and θ to satisfy $\alpha = 1/\eta^2$ and $\theta = \mu\eta^2$. The values that we used for μ and η for L and G are discussed below.

As mentioned above we used the Γ distribution as a parent distribution because it produces values for L and G that are always positive. The Γ distribution has a single peak with a tail at high values, and it approaches a Gaussian distribution as α becomes large (i.e., as η^2 approaches zero). We wished to generate values of $\eta(G)$ which were relatively

large ($\eta^2(G) \sim 0.2$ and $\eta(G) \sim .45$), and if we had used a Gaussian distribution for G we would have occasionally generated non-physical negative values.

For the Γ distribution that described G , we set $\mu_G=1$ and $\eta_G=0.45$. This value for η_G gives $\eta^2(G)=0.2025$, which is comparable to what we reported in the main text for $\eta^2(G)$. For the case that a constitutive promoter drove either reporter, we chose $\mu_L=1$ and $\eta_L=0.059$. This gives $\eta^2(L)=0.0035$, which is one of the larger values that we measured for the pathway variability associated with the constitutive promoters we analyzed.

For the case of the α factor sensitive promoter P_{PRM1} driving YFP , we allowed a negative correlation between L_y and G . We introduced this negative correlation into our Γ -distributed random variables as follows. For a given simulated cell i we first chose a random value for G_i , as described above. Using this value, G_i , we then calculated the quantity $\Delta G_i = G_i - \mu_G$, which describes how far G_i is from the mean value of G . If the simulated cell had ΔG_i positive, then, for that cell i , we reduced the mean value of the parent distribution for pathway power, $\mu_{L,PRM1}$. Similarly if ΔG_i was negative, we increased $\mu_{L,PRM1}$. In this way we introduced a negative feedback into the model: cells with higher G had lower values of L_y on average, and cells with lower G had higher values of L_y on average. Specifically, for cell i , we calculated the mean of the Γ distribution for L_y as

$$\mu_{L,PRM1} = \exp(\beta \Delta G_i) \quad (43)$$

where “ $\exp()$ ” is the exponential function. The parameter β determines the size and sign of the correlation. Thus, if β is negative, then we effectively introduce a negative feedback, where cells with larger values of G_i (ie, ΔG positive) produce smaller values of the mean $\mu_{L,PRM1}$, and therefore smaller values of L_i on average. To calculate the correlation coefficient, $\rho(L, G)$, between L and G we produced a two-dimensional scatter

plot of L vs G and calculated the correlation coefficient using the methods described in section 6.2 above.

We distinguished two regimes for the case of an α factor sensitive promoter driving *YFP*: low and high doses of α factor. For low doses, we generated L_y with the parameters $\eta_L=0.22624$ and $\beta=-0.3$; and for high doses we used $\eta_L=0.059$, which is the same value we used for the constitutive promoters, and $\beta=-0.075$. Both of these combinations of η_L and β give a correlation coefficient $\rho(L_y, G)=-0.48$. This is the value for $\rho(L_y, G)$ that we used in the right hand side of equation (37).

For the constitutive promoters driving either YFP or CFP, we did not simulate any correlation between G and L . The values for η_L for the constitutive promoters are small enough that any potential correlation is small. Thus, when calculating the right hand sides of equations (37) and (38) for constitutive promoters, we set $\rho(L, G)$ to zero.

Finally, for the stochastic contribution to gene expression γ we used a Gaussian distribution of mean zero and width $\eta^2(\gamma)$. Since $\eta^2(\gamma)$ is small, we did not need to worry that the sum $G_i + \gamma_i$, that appears in equation (40) and (41), would be negative. For every simulated cell, we set the transmission noise λ_i to zero, treating the transmission noise as if its contribution to cell-to-cell variation was subsumed in cell-to-cell differences in pathway power L .

We show the results of our simulation in Table S2 for different combinations of promoter pairs: P_{ACT1} driving both CFP and YFP, P_{PRM1} driving both CFP and YFP at low and high α factor, and P_{ACT1} driving CFP with P_{PRM1} driving YFP at low and high α factor. For each of the quantities $\eta^2(y)$, $\eta^2(c)$, and $Z(y, c)$, the table shows three numbers. The first number is the value that we obtained from our simulated data as described above. This number is not affected by the linearization approximation. We note that the value for this first number for $\eta^2(y)$, $\eta^2(c)$, and $Z(y, c)$ are all comparable to what we have

observed in the data. The second number is the value that we calculated using the simulation values of $\eta^2(L)$, $\eta^2(G)$, $\rho(L,G)$ and $\eta^2(\gamma)$ according to the right hand sides of equations (37), (38), and (39). These values include the effect of the linearization approximation. The third number is the percentage difference between the first two. This is the percentage error that we have introduced into our results by making the linearization approximation.

For each of the quantities, the statistical uncertainty associated with the size of the generated sample is $\sim 2.5\%$. This number is significantly smaller than the statistical uncertainty associated with the size of the real data samples ($\sim 10\%$) and therefore, we did not need to generate more simulated data. The approximation that we have made by omitting the higher order terms in equation (15) caused our interpretation of the quantities $\eta^2(c)$, $\eta^2(y)$, and $Z(y,c)$ to be incorrect by the amounts we show in Table S2 (see Figure 1a of main text and equations (22), (27), and (29) of this document). For $\eta^2(y)$ and similarly for $\eta^2(c)$ the table shows that the right hand side of equation (22) is incorrect by a negligible amount. On the other hand, the right hand side of equations (27) and (29), which describe our interpretation of $Z(y,c)$ are incorrect by a more significant percentage, $\sim 10\%$ to 15% . This error is comparable to the statistical uncertainties on $Z(y,c)$ in the real data. We measured the gene expression noise and the pathway variation using $Z(y,c)$, and, therefore, our results here indicate that these values have a systematic uncertainty associated with them of around 10 to 15%.

This Monte Carlo calculation produced surprisingly small errors on the model for $\eta^2(y)$ and $\eta^2(c)$. To investigate this further, we considered a different way of deriving equation (37), where, for the sake of simplicity, we limited ourselves to the YFP case. We started by taking the logarithm of both sides of equation (1) and again, for the sake of simplicity, we neglected the stochastic terms λ and γ . We obtained

$$\ln y_i = \ln L_i + \ln G_i \quad (44)$$

Calculating the variance of both sides, we got, *without approximation*,

$$\sigma^2(\ln y) = \sigma^2(\ln L) + \sigma^2(\ln G) + 2\sigma(\ln L)\sigma(\ln G)\rho(\ln L, \ln G) \quad (45)$$

In general the variance of the logarithm of a variable X is given by

$$\sigma^2(\ln X) = \left(\frac{\sigma(X)}{\bar{X}} \right)^2 + \delta_x = \eta^2(X) + \delta_x \quad (46)$$

where δ_x is a correction that is of higher order than η^2 and where, as always, we have written the variance divided by the mean squared as η^2 . Equation (45) then becomes

$$\eta^2(y) + \delta_y = \eta^2(G) + \eta^2(L) + 2\rho(L, G)\eta(L)\eta(G)\rho(L, G) + (\delta_G + \delta_L + \delta_{\text{cov}}) \quad (47)$$

where the terms δ_y , δ_G , and δ_L are the correction factors of equation (46) for the variables y , G and L respectively. We also introduced the term δ_{cov} for a correction to the correlation coefficient term. On the other hand equation (22) gives

$$\eta^2(y) = \eta^2(G) + \eta^2(L) + 2\rho(L, G)\eta(L)\eta(G) \quad (48)$$

for the case we are considering here of γ and λ set to zero. Comparing equations (47) and (48), we found that the quantity $\delta_G + \delta_L + \delta_{\text{cov}} - \delta_y$ is the error on equation (48) that is produced by the linearization approximation (for the case of γ and λ set to zero). We thus obtain the result that equation (48) would be exactly correct if $\delta_y = \delta_G + \delta_L + \delta_{\text{cov}}$. The small percentage error estimated by the Monte Carlo method between the left hand side and right hand side of equation (37) (and by extension equation (48)) indicates δ_y and $\delta_G + \delta_L + \delta_{\text{cov}}$ have similar values. In other words, the sum $\delta_G + \delta_L + \delta_{\text{cov}}$ can be considered to be a model for the term δ_y . This is not unexpected, perhaps, because L_i and G_i are used to calculate y_i .

7.2 Statistical Uncertainties and the Bootstrap

The uncertainties on all quantities derived from the covariance matrix were calculated using a bootstrap technique¹³. This technique consists of resampling the data with replacement (we chose a cell from the data at random for the resampled data, and we allowed the same cell to be chosen again). Therefore, after each resampling we ended up with a second sample the same size as the original data, and with cells potentially included multiple times. We then used this second scatter plot to calculate the covariance matrix again. We did this 100 times, and kept track of the results of all calculated quantities for the 100 resamples. The standard deviations of the 100 samples then gave us an estimator of the standard error on the different quantities. The bootstrap gave us an easy way to calculate the standard error on quantities that are potentially complicated functions of correlated variables. It also allowed us to account for any additional statistical uncertainty introduced by our method of calculating the covariance matrix.

For plots involving time courses, we wish to know that part of the uncertainty at each time point that is uncorrelated to the previous time point. The uncertainties calculated as above would give the total uncertainty at each time point. To calculate the uncorrelated parts of the uncertainties we also made use of the bootstrap technique, as follows. We resampled the cells and then re-calculated all the covariance matrices for the entire time course. We recorded the results for all quantities that we derived from the covariance matrices for every resampling and for every time point. For each quantity we made a scatter plot of that quantity at time point i with that quantity at the previous time point, where the scatter plot is for the 100 resamplings. We calculated the correlation coefficient $\rho(i, i-1)$ between time point i and the previous time point, $i-1$. That part of the variance σ_i^2 that can be explained by the correlation with the previous time point is given by $\sigma_i^2 \times \rho^2(i, i-1)$. We considered the remaining part of the variance to be the “uncorrelated” part of the variance, and the statistical uncertainty that we associated with time point i was then given by

$$\sqrt{\sigma_i^2 \times (1 - \rho^2(i, i-1))} \quad (49)$$

where σ_i is the standard deviation of the 100 bootstrap resamplings for the quantity in question at time i . For the case that the two time points are 100% correlated, equation (49) would produce no statistical uncertainty on the difference, which is what we expect for 100% correlation; and for the case of 0% correlation, equation (49) reduces to the previous result for the standard error calculated from the bootstrap.

7.3 Contribution of Experimental Error

We consider two classes of experimental error. One class increases the correlation between the measured YFP and CFP signals and the other affects the two colors independently. The fractional errors on the YFP and CFP signals are given by

$$\epsilon_y = \epsilon_{cor} + \delta_y \quad (50)$$

$$\epsilon_c = \epsilon_{cor} + \delta_c \quad (51)$$

where the part of the error that is in common for both CFP and YFP is given by ϵ_{cor} , and the uncorrelated part of the errors are given by δ_y and δ_c . For example, errors in measuring the cell volume will scale both the CFP and YFP signals equally. This is a correlated error and will go into ϵ_{cor} . Similarly, if there is a variation in some property (for example, intracellular pH) among the cells, and this property affects both color variants equally, then this effect will be subsumed in ϵ_{cor} . If there is a variation in this property among the cells and it affects the colors differently, then we will have an uncorrelated error. This uncorrelated error will contribute to δ_y and δ_c .

The measured YFP and CFP differ from the “true” quantities according to

$$y_{measured} = y_{true} \times (1 + \epsilon_{cor} + \delta_y) \quad (52)$$

$$c_{measured} = c_{true} \times (1 + \epsilon_{cor} + \delta_c) \quad (53)$$

We recalculate the total variation in YFP and CFP and also the quantity $Z(y,c)$. We neglect second order terms in ϵ and also products of ϵ and the fractional deviations from the means (η). We get

$$\begin{aligned}\eta_{measured}^2(y) &= \eta_{true}^2(y) + \overline{\epsilon_{cor}^2} + \overline{\delta_y^2} \\ \eta_{measured}^2(c) &= \eta_{true}^2(c) + \overline{\epsilon_{cor}^2} + \overline{\delta_c^2} \\ Z_{measured}(y,c) &= Z_{true}(y,c) + \frac{1}{2}(\overline{\delta_y^2} + \overline{\delta_c^2})\end{aligned}\tag{40}$$

We see that both the correlated and uncorrelated parts affect the total variation, while $Z(y,c)$ is only dependent on the uncorrelated part.

The gene expression noise calculation only used the quantity $Z(y,c)$, and so all our gene expression noise values were too high by the quantity $\frac{1}{2}(\overline{\delta_y^2} + \overline{\delta_c^2})$. We fit $\eta^2(\gamma)$ from the data to the function $\eta^2(\gamma) = A + B/\mu$ where μ was the average cellular total YFP fluorescence as discussed in section 5.3 above. The quantity $\frac{1}{2}(\overline{\delta_y^2} + \overline{\delta_c^2})$ would contribute to the constant term A . We consider the possibility that the constant term is an experimental artifact unrelated to actual gene expression. For example, as mentioned above, this term may be the result of the cell population having a variation in intracellular pH that affects YFP and CFP differently. The resulting non-correlation would persist even in the case of exactly equal numbers of YFP and CFP molecules, or, similarly, in the limit of very large numbers of YFP and CFP molecules.

We calculated $\eta^2(P)$ by calculating differences of values for $Z(y,c)$ from different experiments. If we assume that the average of δ_y^2 and δ_c^2 are the same for all experiments, then these terms do not affect our calculations of transmission noise. This is apparent from the calculations of A_1 , A_2 , and A_3 in equation (30). We used A_1 , A_2 , and A_3 to calculate $\eta^2(P)$, and since A_1 , A_2 , and A_3 are not affected by the experimental errors, the transmission noise values are not either. We thus have the result that the transmission

noise calculation is insensitive to experimental errors to the first order of the fractional errors ϵ .

The calculation of expression capacity variation as given in equation (34) contains the difference of the total variation and $\eta^2(\gamma)$ and $\eta^2(P)$. But $\eta^2(P)$ is insensitive to the fractional errors and $\eta^2(\gamma)$ is calculated from $Z(y,c)$, and so we expect the uncorrelated part of the uncertainty on $\eta^2(y)$ to cancel out. We get that the measured value for $\eta^2(E) + 2\rho(L,E)\rho(L)\rho(E)$ using equation (34) is too high by $\overline{\epsilon_{cor}}^2 + \frac{1}{2}(\overline{\delta_y^2} - \overline{\delta_c^2})$.

Similarly, if we use equation (35), we get that the measured value for $\eta^2(E) + \rho(L_y,E)\eta(L_y)\eta(E) + \rho(L_c,E)\eta(L_c)\eta(E)$ is too high by $\overline{\epsilon_{cor}}^2$. The error in this case will be independent of the uncorrelated errors.

8 Protein Methods

8.1 Sample collection

We collected 10 ml of cells (O.D. ₆₀₀ ~ 0.6) by filtration. Then, we resuspended the cells from the filter in 1 ml of cold YAPD and transferred them to microfuge tubes, pelleted them by centrifugation, aspirated off the supernatant and froze the pellets in liquid N₂. The key to seeing the maximal amount of phosphorylation was the collection of the cells by filtration, as opposed to centrifugation of 10 ml of culture.

8.2 Cell extracts

We prepared total cellular extracts of denatured proteins according to the method of Yaffe and Schatz¹⁴. Briefly, we resuspended a pellet in 300 μ l of 1.85M NaOH, 7.4% β -mercaptoethanol and incubated for 10 min on ice, then added 300 μ l of 50% (w/v)

trichloroacetic acid and continued incubation for 10 min on ice. We pelleted the precipitate by centrifugation in a 4°C microfuge for 2 min at maximum speed, washed twice with 1 ml acetone at –20°C, dried briefly, and resuspended in gel loading buffer (30 mM Tris base, 50 mM Tris-HCl (pH 6.8), 1.25 mM EDTA, 50 mM DTT, 1% SDS, 10% glycerol, 0.1% bromophenol blue). We assayed protein concentrations of cell extracts using the Pierce Micro-BCA assay kit and BSA as protein concentration standard.

8.3 Western Blot Protocol

We loaded 50 µg of cell extract per lane on a 4-15% gradient gel (BioRad) and electrophoresed at 25 mM . We electroblotted the proteins from the gel onto a PVDF membrane (Immobilon, Millipore) in cold transfer buffer for 40 min at 220 mA. After blotting, we blocked the membrane in TBS/1% casein, and then hybridized overnight at 4°C in TBST/1% casein and 1/1000 anti-phospho p44/42 (Thr202/Tyr204) (rabbit polyclonal, #9101L, Cell Signaling Technologies). After washing 3x 10 min in TBS at room temperature, the membrane was hybridized with an anti-rabbit peroxidase-labeled secondary antibody (1/150,000, #31460, Pierce) for 1 hr at room temperature in TBS/0.5% casein. After washing 3x 10 min in TBS at room temperature, we visualized the secondary antibodies using the Pierce SuperSignal ECL Western Blotting Detection Reagents (#34075) and Kodak x-ray film. We performed multiple exposures in order to ensure that the signal observed was within the linear range of the film. The second key to seeing the low levels of phosphorylation was the overnight hybridization with the primary antibody.

Supplementary Tables

Table S1. Strains

Strain Name	Relevant Genotype
ACLY379	MATa bar1Δ
ACLY387	MATa bar1Δ Δprm1::P _{PRM1} -YFP::his5 ⁺
ACLY394	MATa bar1Δ cdc28-as2 Δprm1::P _{PRM1} -YFP::his5 ⁺
TCY3096	MATa bar1Δ cdc28-as2 Δprm1::P _{PRM1} -YFP::his5 ⁺ trp1::pP _{PRM1} -CFP-TRP1
TCY3161	MATa bar1Δ P _{BMH2} ::pP _{BMH2} -CFP-TRP1 ura3::pP _{BMH2} -YFP-URA3
TCY3162	MATa bar1Δ P _{ACT1} ::pP _{ACT1} -CFP-TRP1 ura3::pP _{ACT1} -YFP-URA3
TCY3165	MATa bar1Δ P _{STE5} ::pP _{STE5} -CFP-TRP1 ura3::pP _{STE5} -YFP-URA3
TCY3152	MATa bar1Δ P _{ACT1} ::pP _{ACT1} -CFP-TRP1 ura3::pP _{BMH2} -YFP-URA3
TCY3153	MATa bar1Δ P _{ACT1} ::pP _{ACT1} -CFP-TRP1 ura3::pP _{STE5} -YFP-URA3
TCY3203	MATa bar1Δ P _{BMH2} ::pP _{BMH2} -CFP-TRP1 ura3::pP _{STE5} -YFP-URA3
TCY3146	MATa bar1Δ cdc28-as2 leu2::pP _{PRM1} -CFP-LEU2 ura3::pP _{BMH2} -YFP::URA3
TCY3148	MATa bar1Δ cdc28-as2 leu2::pP _{PRM1} -CFP-LEU2 ura3::pP _{STE5} -YFP::URA3
TCY3204	MATa bar1Δ cdc28-as2 leu2::pP _{PRM1} -CFP-LEU2 ura3::pP _{ACT1} -YFP::URA3
TCY3154	MATa bar1Δ cdc28-as2 Δprm1::P _{PRM1} -YFP-his5 ⁺ trp1::pP _{ACT1} -CFP::TRP1
TCY3248	MATa bar1Δ cdc28-as2 P _{ACT1} ::pP _{ACT1} -CFP-TRP1 ura3::pP _{BMH2} -YFP::URA3
GPY3262	MATa bar1Δ cdc28-as2 Δfus3::kanMX4 Δprm1::P _{PRM1} -YFP::his5 ⁺ trp1::pP _{ACT1} -CFP-TRP1
GPY3263	MATa bar1Δ cdc28-as2 Δkss1::kanMX4 Δprm1::P _{PRM1} -YFP::his5 ⁺ trp1::pP _{ACT1} -CFP-TRP1
TCY3261	MATa bar1Δ P _{ACT1} ::pP _{PRP22} -CFP-TRP1 ura3::pP _{PRP22} -YFP-URA3
TCY3257	MATa bar1Δ P _{ACT1} ::pP _{ACT1} -CFP-TRP1 ura3::pP _{PRP22} -YFP-URA3

Strains used during this work. All were derived from W303a as described in Experimental Procedures and the Supplementary Materials

Table S2. Error Introduced by Model Approximations

Pro- moter combi- nations CFP- YFP	$\eta^2(c)$			$\eta^2(y)$			$Z(y,c)$		
	Gener- ated	Expect from model	Error (%)	Gener- ated	Expect from model	Error (%)	Gener- ated	Expect from model	Error (%)
ACT1- ACT1	0.2177	0.2167	-0.5%	Same as $\eta^2(c)$			0.00255	0.00228	-10.5%
PRM1- PRM1	0.1737	0.1764	1.5%	Same as $\eta^2(c)$			0.01351	0.01159	-14.2%
Low α PRM1- PRM1	0.1890	0.1928	2.0%	Same as $\eta^2(c)$			0.00255	0.00239	-6.3%
High α ACT1- PRM1	0.2128	0.2154	1.2%	0.1737	0.1764	1.5%	0.04918	0.04380	-10.9%
Low α ACT1- PRM1	0.2198	0.2177	1.0%	0.1853	0.1886	1.8%	0.00756	0.00644	-14.8%
High α									

Results of Monte Carlo calculations described in section 7.1. The column labelled “Error” is the percentage error introduced by making the approximations in our model discussed in the text. The five cases we show are for a simulation of ACT1 driving CFP and YFP (first row); PRM1 driving CFP and YFP at low pheromone (second row) and high pheromone (third row); and ACT1 driving CFP and PRM1 driving YFP at low (fourth row) and high doses of α factor (fifth row).

Table S3. Contribution of different sources of variation for constitutive promoters

	Gene expression noise, $\eta^2(\gamma)$	Variation in Pathway output, $\eta^2(P)$	Variation in G (+ correlation)
P_{STE5}			0.22±0.03
	0.0268±0.0043	0.0006 ± 0.0080	
P_{BMH2}		0.0013 ± 0.0065	0.21±0.03
	0.00450±0.00052		
P_{ACT1}	0.00204±0.00031	0.0013 ± 0.0046	0.21±0.03
P_{PRP22}	0.0019±0.0027	0.0198 ± 0.011*	0.21±0.03

Distribution of total cell-to-cell variation η^2 between different sources for four constitutive promoters. We treated strains TCY3161, TCY3162, TCY3165, TCY3152, TCY3153, TCY3261, TCY3257 and TCY3203 with 20 nM α factor for 2.5 hours and then collected YFP and CFP images. We calculated the variation due to each of the different sources of variation as explained in the Results section and Supplementary Materials. The errors shown are the standard error. In addition, we associate a 10 to 15% systematic uncertainty with the reported numbers for the variation in pathway output due to the omission of higher order terms in equations 2 and 3 (see Suppl Mat).

* For the calculation of pathway variation for PRP22 we assumed the value for $\eta^2(P)$ for P_{ACT1} listed in the table.

Supplementary Figure Legends

Figure S1. The ACT1 promoter is insensitive to α factor.

We treated TCY3154 cells (P_{PRM1} -YFP P_{ACT1} -CFP) with 20 nM (●), 4nM (▲) or 0.16 nM (■) α factor and 10 μ M Cdc28-as2 inhibitor (1-NM-PP1), and collected images at 15-minute intervals as explained in Materials and Methods and Supplementary Material.

Data corresponds to the mean amount of CFP fluorescence produced during the experiment (mean at a given time minus mean at time 0) $\pm$ standard error in the mean.

Note that production of CFP is the same at different doses of α factor. Also, the increase in the signal over time observed for all doses is not induced by α factor. It is caused by the fact that cells stop dividing at time 0 when Cdc28-as2 inhibitor is added. From then on, all new CFP produced remains in the same cell without being diluted by cell division.

Figure S2. Expression capacity influences the rate at which cells increase in volume.

We treated TCY3154 cells (P_{PRM1} -YFP, P_{ACT1} -CFP) with 20 nM α factor and 10 μ M Cdc28-as2 inhibitor (1-NM-PP1) and collected CFP images at 15-minute intervals. **a.**

Cells increase in volume exponentially when treated with α factor. Here we show examples of 3 cells that at the time of stimulation had similar volumes but grew at different rates. Symbols correspond to the values reported by CELL-ID and the lines correspond to a fit to the exponential curve $V(t) = V_0 e^{gt}$ where V_0 is the fitted volume at time 0, and g is the “fractional growth rate.” **b.** Change in cell volume (ΔV) vs. initial volume (V_0). **c.** Fractional growth rate (g) vs. V_0 using fit results. **d.** Total P_{ACT1} -derived CFP fluorescence produced in each cell during the experiment (ΔCFP) vs. ΔV . **e.** Total amount of P_{ACT1} -derived CFP fluorescence produced during the experiment (ΔCFP) vs. V_0 . In b-e, data correspond to the values obtained after 3 hours of stimulation. **f.** Cellular

concentration of P_{PRM1} -derived YFP fluorescence vs P_{ACT1} -derived CFP fluorescence 3 hours after stimulation. Data corresponds to the total YFP or CFP fluorescence in each cell divided by their volume.

Figure S3. Dose response becomes sharper over time due to the inhibitory effect Cdc28.

We treated *CDC28* (ACL387, left) or *cdc28-as2* (TCY3154, right) cells with the indicated doses of α factor in the absence (left) or presence (right) of 10 μ M Cdc28-as2 inhibitor (1-NM-PP1) and measured α factor system output 1 (○), 2.5 (◆) and 3.5 (■) hours later. Symbols correspond to the median $\pm$ standard error, expressed as percentage of the maximum response at each time point. Lines on the left panel are show to guide the eye. We observed that the dose response became steeper at longer times (left panel) but that this effect was suppressed when we inhibited Cdc28 (right panel). This result suggested that the “sharpening” of the response to α factor is dominated by the cross-inhibition of the α factor pathway and the cell cycle machinery *via* Cdc28. When we cut the connection between the cell cycle and the α factor pathway (by adding inhibitor) we also remove the time-dependent sharpening.

Figure S4. Gene expression noise is low and similar for different promoters.

We treated TCY3096 cells with 10 μ M Cdc28-as2 inhibitor (1-NM-PP1) and 20 nM α factor and collected YFP and CFP images every 15 minutes (●) or with inhibitor and increasing doses of α factor for 2.5 hours (▲). In addition, we treated cells with both YFP and CFP under the control of the same constitutive promoter (P_{ste5} ■ TCY3065, P_{bmh2} ◆ TCY3061 and P_{act1} ▼ TCY3062) with inhibitor and 20 nM α factor for 2.5 hours. Plot shows gene expression noise ($\eta^2(\gamma)$) vs median YFP signal. We fitted gene expression noise to $\eta^2(\gamma)=A+B/\mu$, where μ is the mean YFP signal, $A=0.0015 \pm 0.0001$, and $B=7000 \pm 500$ (black line). Errors bars are estimations of the standard error obtained by the bootstrap method (Supplementary Materials).

Figure S5. Alpha factor does not affect the correlation between two constitutive promoters.

We treated TCY3248 cells (P_{BMH2} -YFP, P_{ACT1} -CFP) with 20 (●) or 1.25 nM (●) α factor and 10 μM Cdc28-as2 inhibitor (1-NM-PP1) and collected YFP and CFP images every 15 minutes. Plot shows the output of the α factor response system vs. ACT1 system 3 hours after stimulation. We normalized YFP and CFP F.U. by the median of each population to allow overlaying data with different means. Note that, in contrast to the results shown in Figure 5a for the P_{PRM1} -YFP reporter, the uncorrelated variation (dispersion along the minor axis) does not increase at low doses of α factor.

Figure S6. Contribution of different sources of variation change with α factor dose.

Relative contribution to overall cell-to-cell variation of different sources in cells treated with 1.25 nM (a) or 20 nM (b) α factor and 10 μM Cdc28-as2 inhibitor (1-NM-PP1) as in Figure 5. We show variation of P_{ACT1} -derived CFP (total η^2 , ✕) and of P_{PRM1} -derived YFP (total, ◆; expression noise $\eta^2(\gamma)$, ■; pathway variation $\eta^2(P)$, ▲; and expression capacity variation plus correlation $\eta^2(Gp)$, ▼). The error bars on each point indicate the standard error between time points, while the vertical bars with no symbol indicate the standard error associated with repetitions of the experiment (see Suppl Mat).

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